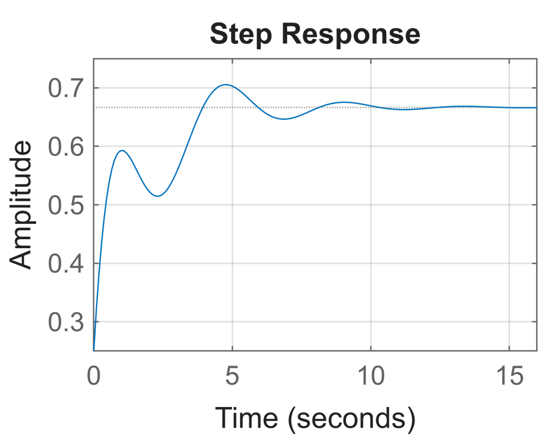
```
%L2.a. State Space Analysis

% Problem 1
% Aim: To represent transfer function in state space form and vice versa
% Also to obtain step response of SISO system
% Consider a plant transfer function
% G(s)=(s^3+5s^2+4s+6)/(4s^3+7s^2+12s+9)

% Transfer function coefficients
num1=[1 5 4 6];
den1=[4 7 12 9];
% Transfer function to state space conversion
[A,B,C,D]=tf2ss(num1,den1)
```

```
A = 3×3
   -1.7500   -3.0000   -2.2500
    1.0000         0         0
         0    1.0000         0
B = 3×1
     1
     0
     0
C = 1×3
    0.8125    0.2500    0.9375
D =
0.2500
```

```
% State space to transfer function conversion
[num3,den3]=ss2tf(A,B,C,D)
```

```
num3 = 1×4
    0.2500    1.2500    1.0000    1.5000
den3 = 1×4
    1.0000    1.7500    3.0000    2.2500
```

```
% Step response from state space model
figure(1)
step(A,B,C,D)
grid on
title('Step Response')
```



```
%L2.a. State Space Analysis

% Problem 2
% MIMO system time domain response

% State space matrices
A=[0 1;-25 -4];
B=[1 1;0 1];
C=[1 0;0 1];
D=[0 0;0 0];

% Transfer functions corresponding to input u1
[num1,den1]=ss2tf(A,B,C,D,1)
```

```
num1 = 2×3
     0     1     4
     0     0   -25
den1 = 1×3
    1.0000    4.0000   25.0000
```

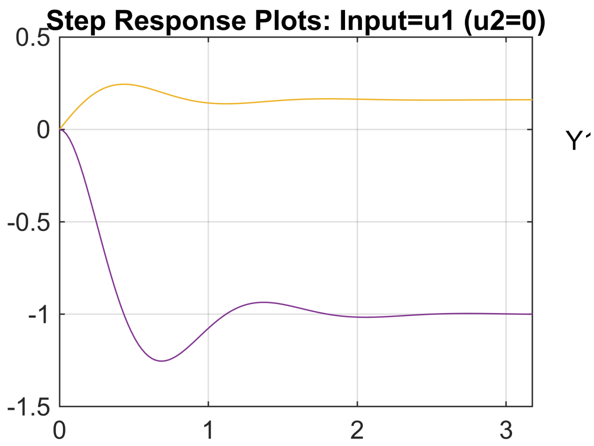
```
% Transfer functions corresponding to input u2
[num2,den2]=ss2tf(A,B,C,D,2)
```

```
num2 = 2×3
         0    1.0000    5.0000
         0    1.0000  -25.0000
den2 = 1×3
    1.0000    4.0000   25.0000
```

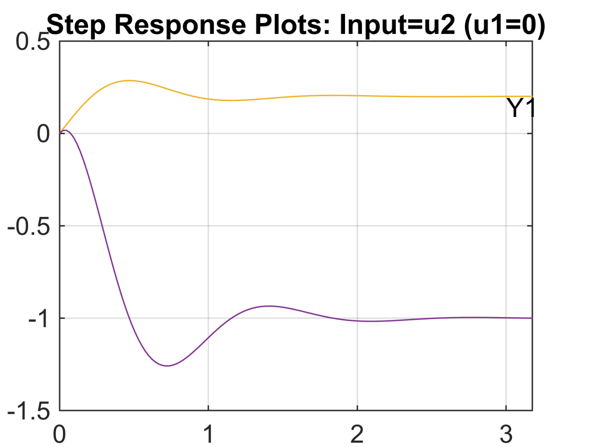
```
% Eigenvalues of A matrix
eig(A)
```

```
ans = 2×1 complex
  -2.0000 + 4.5826i
  -2.0000 - 4.5826i
```

```
% Step response for input u1 (u2=0)
figure(2)
[y1,y2,t]=step(A,B,C,D,1);
plot(t,y1,t,y2)
grid on
title('Step Response Plots: Input=u1 (u2=0)')
text(3.4,-0.06,'Y1')
text(3.4,1.4,'Y2')
```




```
% Step response for input u2 (u1=0)
figure(3)
[y1,y2,t]=step(A,B,C,D,2);
plot(t,y1,t,y2)
grid on
title('Step Response Plots: Input=u2 (u1=0)')
text(3,0.14,'Y1')
text(2.8,1.1,'Y2')
```




```
%L2.a. State Space Analysis

% Problem 3
% Diagonalization of A matrix in Controllable Canonical Form
% Determine Jordan canonical form and verify system properties

% State space matrices
A=[0 1 0;0 0 1;-6 -11 -6];
B=[0;0;2];
C=[1 0 0];
D=[0];
% Eigenvalues of A
L=eig(A)
```

```
L = 3×1
   -1.0000
   -2.0000
   -3.0000
```

```
% Van der Monde transformation matrix
P=[1 1 1;
   L(1) L(2) L(3);
   L(1)^2 L(2)^2 L(3)^2];
% Inverse of transformation matrix
PI=inv(P);
% Diagonalized state model
ASTILDA=PI*A*P
```

```
ASTILDA = 3×3
   -1.0000   -0.0000   -0.0000
   -0.0000   -2.0000   -0.0000
    0.0000   -0.0000   -3.0000
```

```
BSTILDA=PI*B
```

```
BSTILDA = 3×1
    1.0000
   -2.0000
    1.0000
```

```
CSTILDA=C*P
```

```
CSTILDA = 1×3
     1     1     1
```

```
DSTILDA=D
```

```
DSTILDA =
0
```

```
% Eigenvalues of transformed matrix
L1=eig(ASTILDA)
```

```
L1 = 3×1
   -1.0000
   -2.0000
   -3.0000
```

```
% Determinants
DETA=det(A)
```

```
DETA =
-6
```

```
DETSTILDA=det(ASTILDA)
```

```
DETSTILDA =
-6.0000
```

```
% Traces
TRA=trace(A)
```

```
TRA =
-6
```

```
TRASTILDA=trace(ASTILDA)
```

```
TRASTILDA =
-6.0000
```

```
% Transfer function from original state model
[num,den]=ss2tf(A,B,C,D);
S1=tf(num,den)
```

```
S1 =

            2
  ----------------------
  s^3 + 6 s^2 + 11 s + 6

Continuous-time transfer function.
Model Properties
```

```
% Transfer function from transformed state model
[num1,den1]=ss2tf(ASTILDA,BSTILDA,CSTILDA,DSTILDA);
S2=tf(num1,den1)
```

```
S2 =

            2
  ----------------------
  s^3 + 6 s^2 + 11 s + 6

Continuous-time transfer function.
Model Properties
```